Demo

Health and Retirement Study (hrs) dataset

Linear Regression Overview

Linear Regression

- Predict a **continuous outcome** (dependent variable) using one or more predictors (independent variables).

- Describe the relationship between **the outcome** and **predictors**.

The **outcome** (dependent variable) must be **continuous**.

Linear regression does not require the outcome to be normally distributed (but it is helpful).

Demo

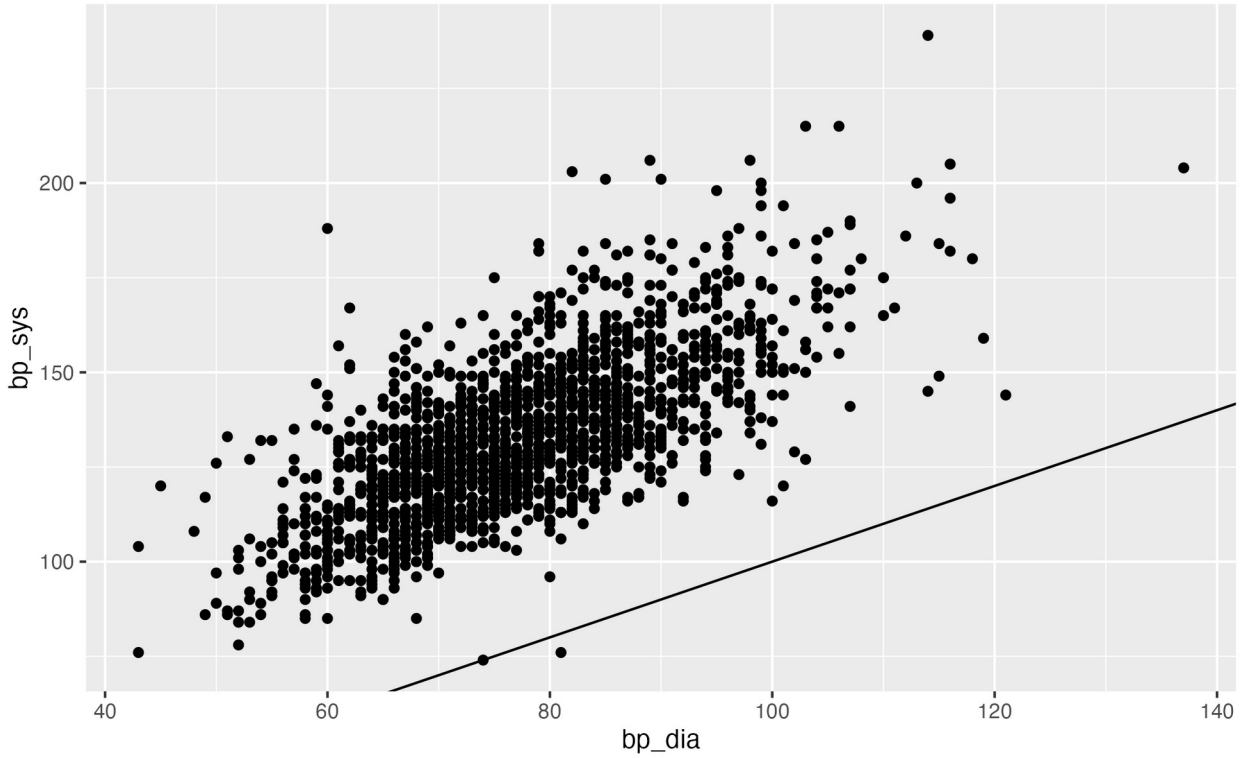
Exploring the data

Knowledge Check

Suppose we are trying to predict systolic blood pressure based on diastolic blood pressure.

True or False: Systolic blood pressure is our predictor variable and diastolic blood pressure is our outcome variable.

Note: the line on this plot is not a linear regression line - it represents where the two variables are equal (for reference only).



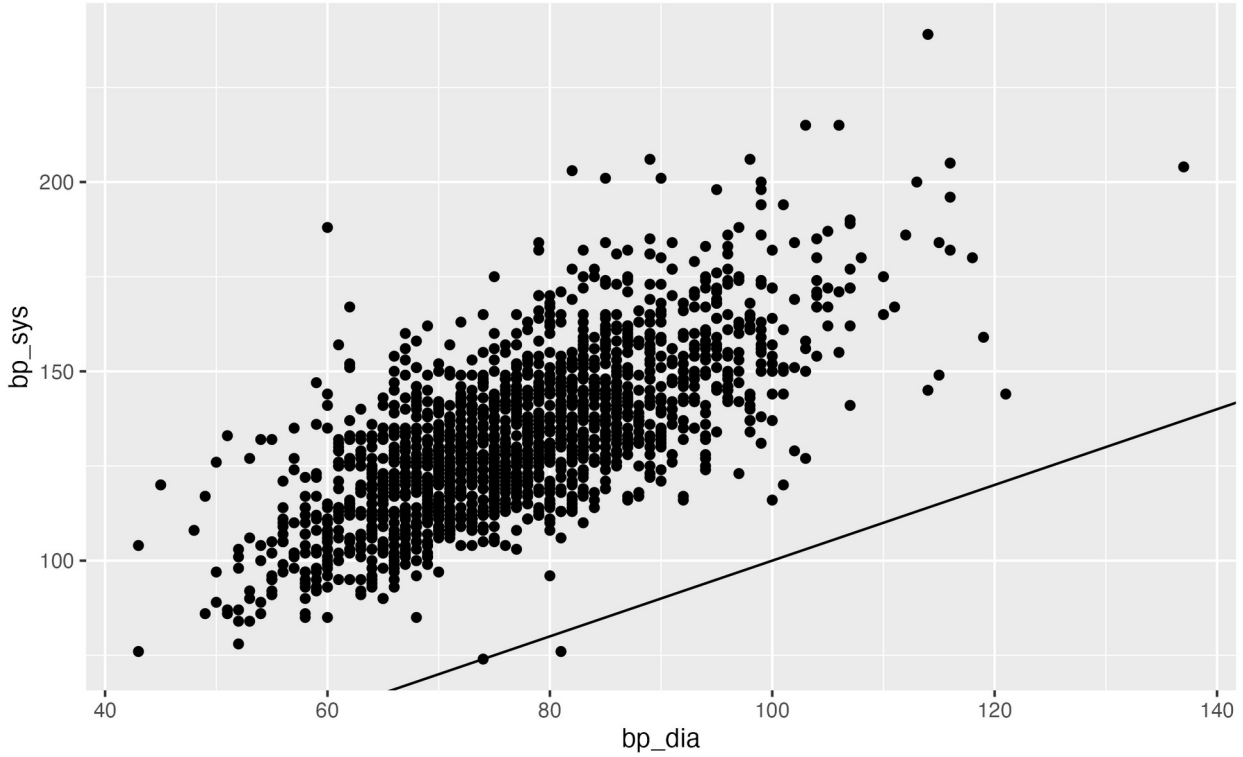
a) True

b) False

Knowledge Check

Suppose we are trying to predict systolic blood pressure based on diastolic blood pressure.

True or False: Systolic blood pressure is our predictor variable and diastolic blood pressure is our outcome variable.



a) True

b) False

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